

CHAPTER 13

WAVES

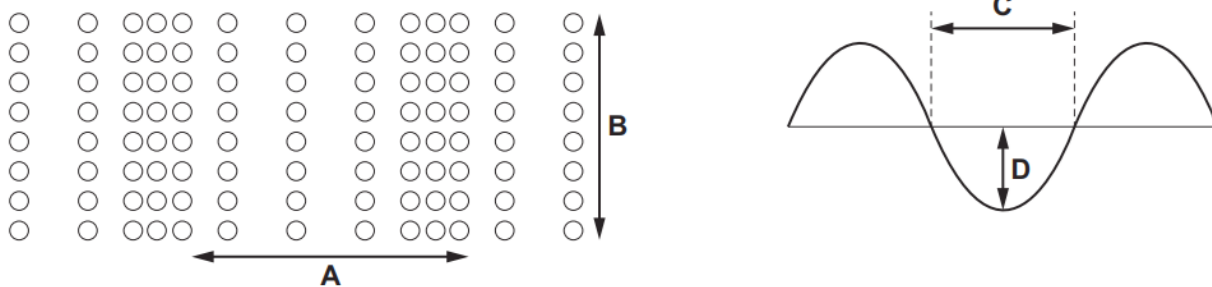
3.1 General properties of waves

- 1 Know that waves transfer energy without transferring matter
- 2 Describe what is meant by wave motion as illustrated by vibrations in ropes and springs and by experiments using water waves
- 3 Describe the features of a wave in terms of wavefront, wavelength, frequency, crest (peak), trough, amplitude and wave speed
- 4 Define the terms:
 - (a) frequency as the number of wavelengths that pass a point per unit time
 - (b) wavelength as the distance between two consecutive, identical points such as two consecutive crests
 - (c) amplitude as the maximum distance from the mean position
- 5 Recall and use the equation
wave speed = frequency $\times$ wavelength
 $v = f\lambda$
- 6 Know that for a transverse wave, the direction of vibration is at right angles to the direction of the energy transfer, and give examples such as electromagnetic radiation, waves on the surface of water, and seismic S-waves (secondary)
- 7 Know that for a longitudinal wave, the direction of vibration is parallel to the direction of the energy transfer, and give examples such as sound waves and seismic P-waves (primary)
- 8 Describe how waves can undergo:
 - (a) reflection at a plane surface
 - (b) refraction due to a change of speed
 - (c) diffraction through a gap
- 9 Describe how wavelength and gap size affects diffraction through a gap
- 10 Describe the use of a ripple tank to show:
 - (a) reflection at a plane surface
 - (b) refraction due to a change in speed caused by a change in depth
 - (c) diffraction due to a gap
 - (d) diffraction due to an edge
- 11 Describe how wavelength affects diffraction at an edge

1. O/N/23/12/Q24

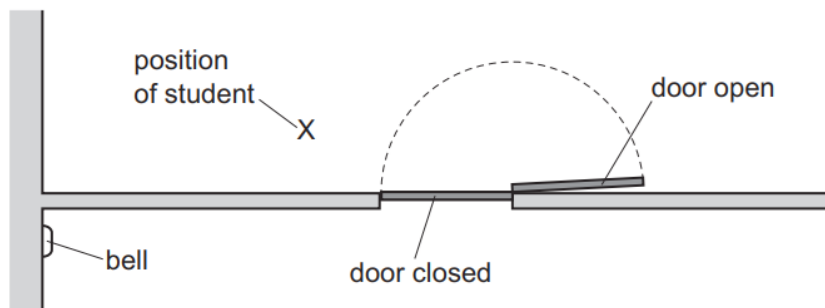
The diagram shows two waveforms.

Which arrow shows the wavelength of a longitudinal wave?



2. O/N/23/12/Q25

A bell, on a wall outside a classroom, rings. A student is standing at point X shown on the view from above.



The door of the classroom is closed and the student in the room at X hears the bell ring.

The door is opened, and the sound heard by the student from the bell is louder.

What causes the sound reaching the student to be louder?

- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

3. O/N/22/11/Q23

Which statement is correct?

- A** Both longitudinal waves and transverse waves can travel at the speed of light.
- B** Both longitudinal waves and transverse waves of similar wavelengths must have similar frequencies.
- C** Both longitudinal waves and transverse waves can travel in a vacuum.
- D** Both longitudinal waves and transverse waves can travel in water.

4. M/J/22/11/Q19

A wave in the sea collides with a cliff.

A crest of the wave hits the cliff once every 6.0 s.

The horizontal distance between a crest and the adjacent trough of the wave is 4.5 m.

What is the speed of the wave?

- A** 0.67 m/s **B** 0.75 m/s **C** 1.3 m/s **D** 1.5 m/s

5. M/J/22/11/Q20

A sound wave refracts as it passes from air to water.

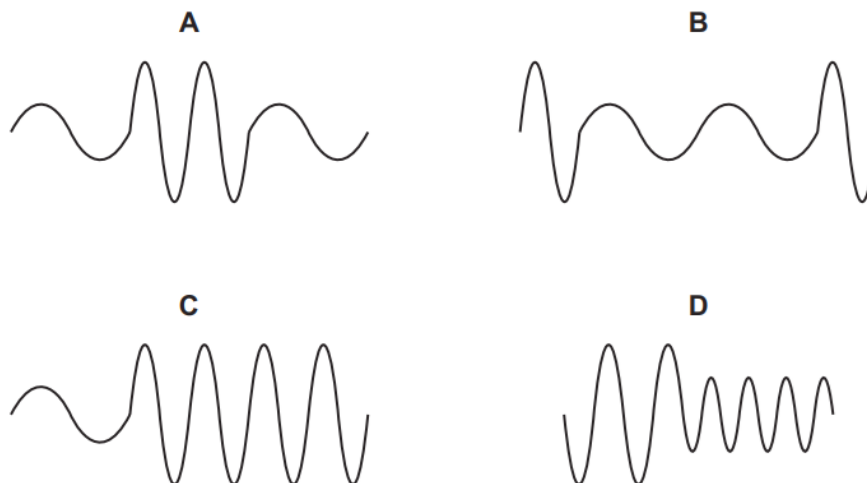
Which quantities change?

- A** speed and frequency only
B speed and wavelength only
C wavelength and frequency only
D speed, frequency and wavelength

6. O/N/21/11/Q17

A water wave on a pond passes into a region where the wave travels more slowly. It then returns to a region where it travels at the original speed. Assume that the frequency of the wave stays constant.

Which side-view of the wave is correct?



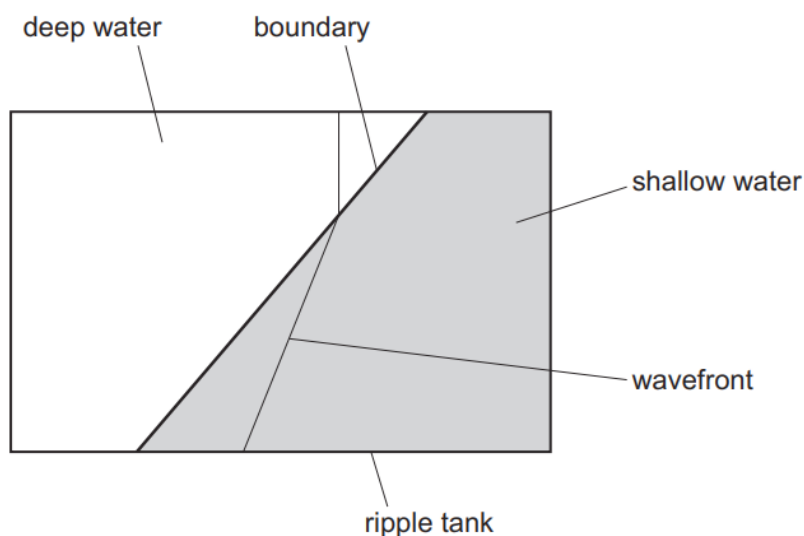
7. O/N/21/11/Q18

What is meant by the *amplitude* of a wave?

- A** the distance between the extreme points of a particle's motion
- B** the maximum distance a particle moves from its equilibrium position
- C** the maximum energy carried by the wave
- D** the maximum power of the vibrations carrying the wave

8. O/N/21/11/Q19

The diagram shows one wavefront of a wave as it travels from deep water to shallow water in a ripple tank.



What happens as the wavefront moves into the shallow water?

- A** The speed of the wavefront increases.
- B** The speed of the wavefront decreases.
- C** The wavelength of the wave remains constant.
- D** The wavelength of the wave increases.

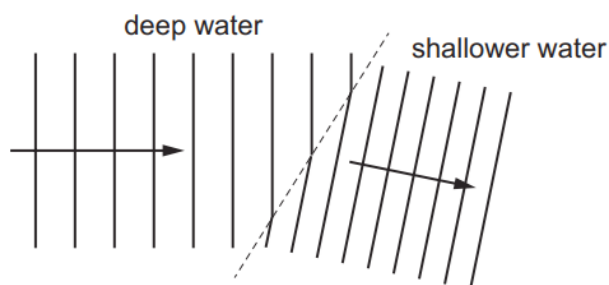
9. O/N/21/12/Q22

Which example describes a transverse wave being refracted?

- A** infrared waves travelling in an optical fibre
- B** light waves from an object entering a magnifying glass
- C** ultrasound waves being used to clean jewellery
- D** water waves hitting a barrier

10. M/J/21/11/Q20

A water wave in a ripple tank refracts as it moves from deep water into shallower water.



What happens to the speed and to the frequency of the wave as it moves into shallower water?

	speed	frequency
A	decreases	stays constant
B	increases	stays constant
C	stays constant	decreases
D	stays constant	increases

11. O/N/20/11/Q26

An electromagnetic wave has a speed of 3.0×10^8 m/s and a wavelength of 10 cm.

What is the frequency of the wave?

- A** 3.3×10^{-10} Hz
- B** 3.3×10^{-8} Hz
- C** 3.0×10^7 Hz
- D** 3.0×10^9 Hz

12. M/J/20/12/Q30

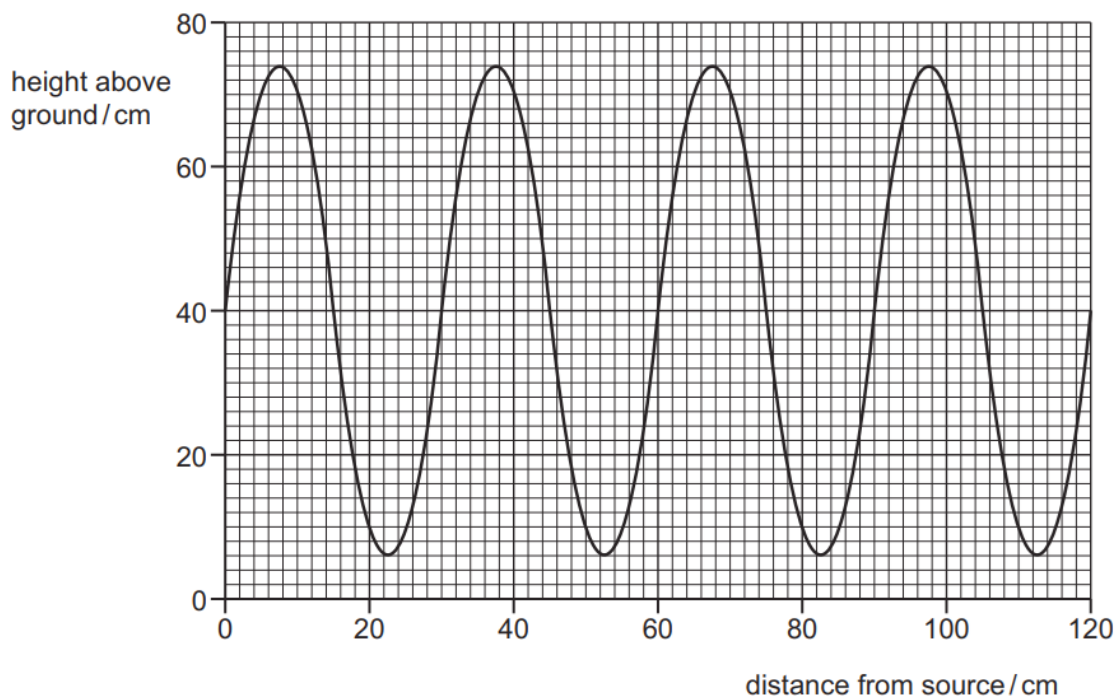
A vibrator produces 12 wavelengths on the surface of water in 10 s. The spacing between the first crest and the third crest is 60 cm.

What is the speed of the wave?

- A** 24 cm/s **B** 25 cm/s **C** 36 cm/s **D** 72 cm/s

13. M/J/20/12/Q29

The diagram shows how the height above the ground of a rope transmitting a wave varies with distance from the source of vibrations at one instant in time.



What is the amplitude of the wave shown?

- A** 30 cm **B** 34 cm **C** 72 cm **D** 74 cm

14. O/N/19/11/Q25

The frequency of microwaves used in an experiment is 3.0×10^9 Hz. Their speed in air is 3.0×10^8 m/s.

What is a possible frequency and a possible speed in air of a radio wave?

	frequency / Hz	<u>speed in air</u> m/s
A	3.0×10^5	1500
B	3.0×10^5	3.0×10^8
C	3.0×10^{14}	1500
D	3.0×10^{14}	3.0×10^8

15. O/N/19/12/Q22

Which description defines the amplitude of a water wave in a ripple tank?

- A** half of the vertical distance between a trough and a peak
- B** the horizontal distance between adjacent peaks
- C** the horizontal distance between a trough and an adjacent peak
- D** the vertical distance between a trough and a peak

16. O/N/19/12/Q23

Here are four statements about wavefronts.

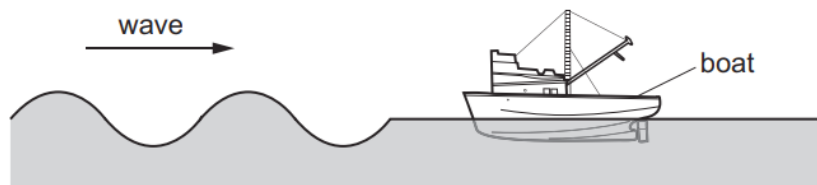
- 1 Wavefronts from a point source are straight lines.
- 2 Wavefronts from a point source are circular.
- 3 Wavefronts are parallel to the direction of energy movement.
- 4 Wavefronts are perpendicular to the direction of energy movement.

Which two statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

17. M/J/19/12/Q25

A wave on water travels towards a stationary boat.



As the wave passes, the boat starts to move up and down and gains kinetic energy from the wave.

Which property of the wave changes as the wave passes the boat?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

18. M/J/19/12/Q26

Which waves are longitudinal?

- A** gamma rays
- B** light waves
- C** ultrasound waves
- D** X-rays

19. O/N/18/11/Q18

Which term is the number of crests of a wave passing a point per second?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

20. O/N/18/12/Q22

Which statement about a water wave is correct?

- A** The amplitude is the vertical distance between a trough and a peak.
- B** The frequency is the number of troughs passing a point in one second added to the number of peaks passing a point in one second.
- C** The speed is the horizontal distance travelled per second by a peak.
- D** The wavelength is the horizontal distance between a trough and a peak.

21. M/J/18/12/Q26

A communications system uses microwaves with a wavelength of 15 cm. The speed of light is 3.0×10^8 m/s.

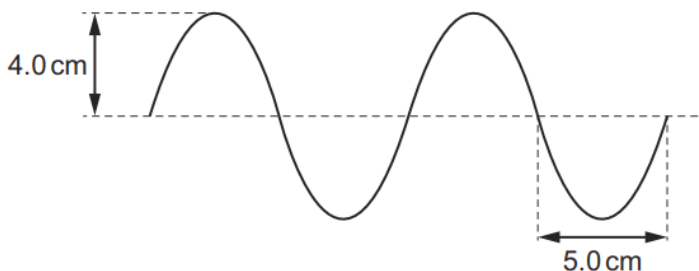
What is the frequency of the microwaves?

- A** 2.0×10^7 Hz **B** 4.5×10^7 Hz **C** 2.0×10^9 Hz **D** 4.5×10^9 Hz

22. O/N/17/11/Q18

A wave is travelling in water in a glass-sided tank.

The diagram shows a side view of the wave at one instant.



What are the amplitude and the wavelength of the wave?

	amplitude / cm	wavelength / cm
A	4.0	5.0
B	4.0	10
C	8.0	5.0
D	8.0	10

23. O/N/17/11/Q19

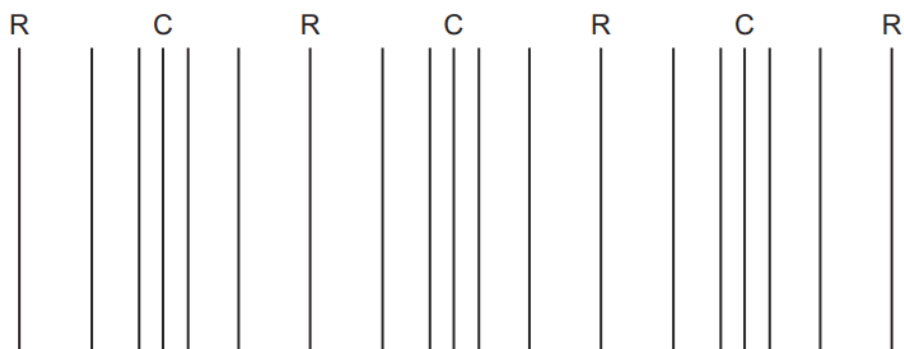
An earthquake wave travels through the solid surface of the Earth from east to west. The solid surface vibrates in a north-south direction.

How can the earthquake wave be described?

- A** electromagnetic
- B** longitudinal
- C** sound
- D** transverse

24. O/N/17/12/Q19

A sound wave in a solid is represented by a series of lines. The diagram shows compressions C and rarefactions R at one instant in time.



Which statement is correct?

- A** All particles on one line move in the same direction at the same time.
- B** The direction of travel of the sound wave is parallel to the lines.
- C** The distance between a compression and its adjacent rarefaction is a wavelength.
- D** The pattern of lines represents a transverse wave.

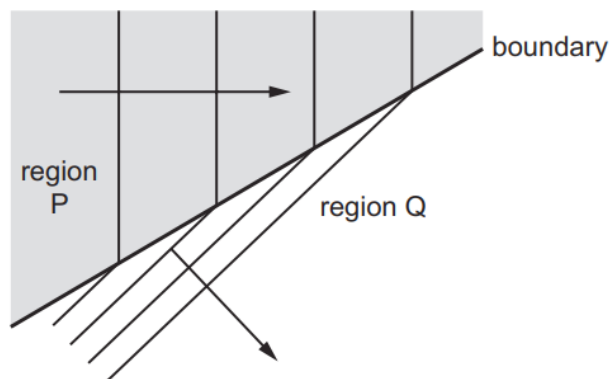
25. M/J/17/11/Q26

Which waves are longitudinal?

- A** sound waves in water
- B** ultra-violet waves in air
- C** waves on the surface of water
- D** X-rays in a vacuum

26. M/J/17/12/Q26

The diagram shows a water wave in a ripple tank being refracted at the boundary between region P and region Q.

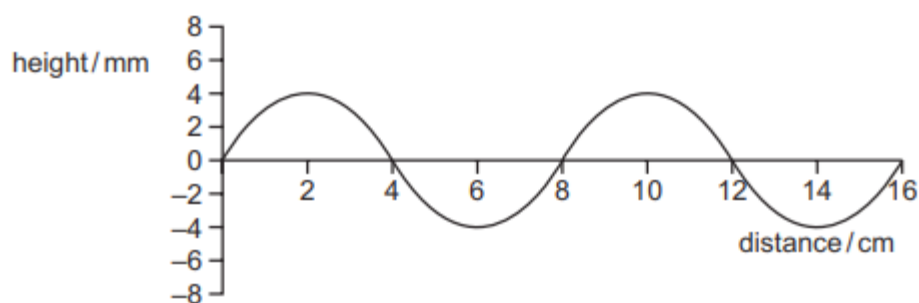


How does the wave in region P compare with the wave in region Q?

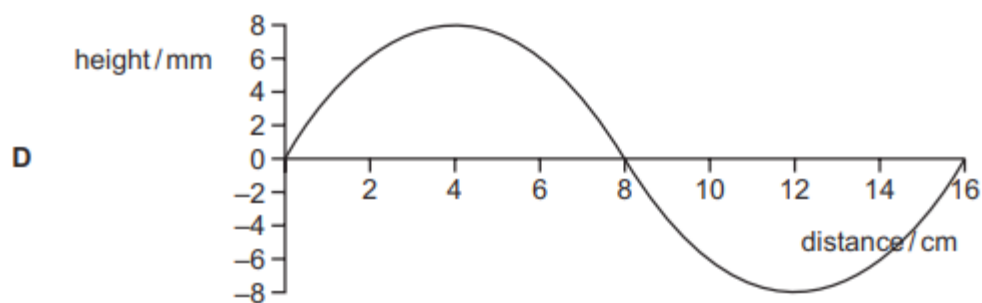
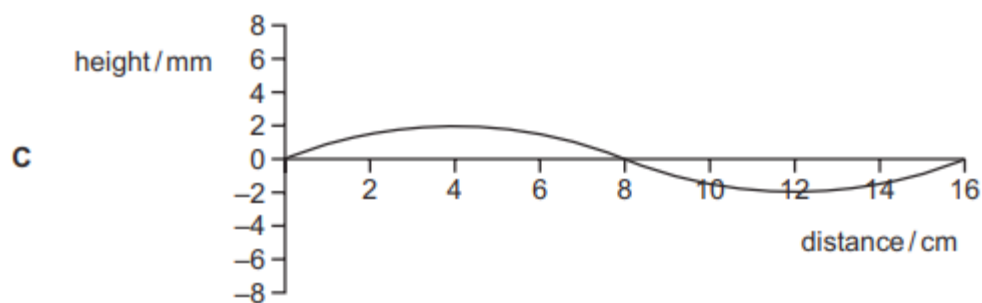
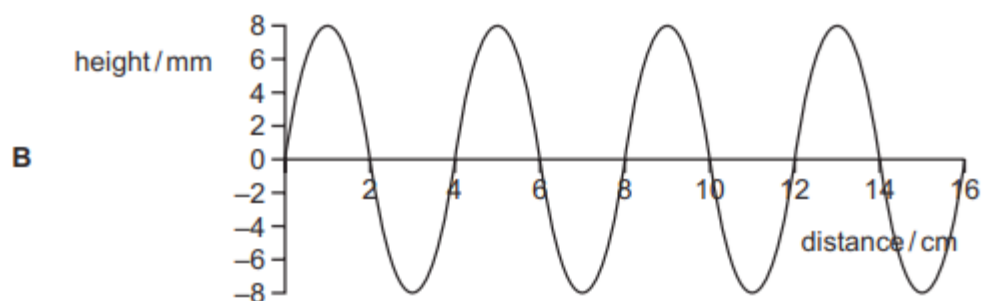
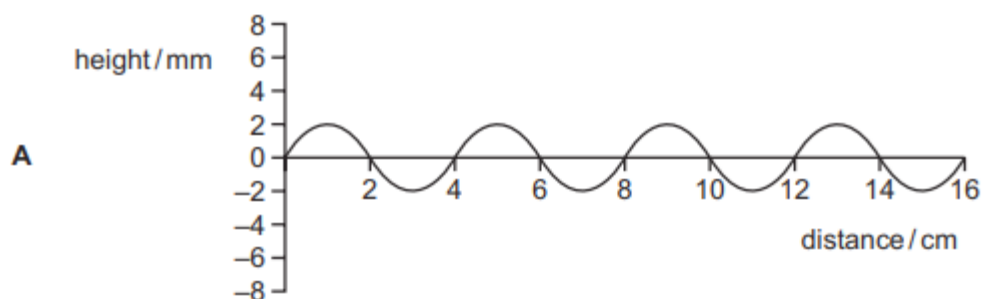
	wavelength of wave in region P	speed of wave in region P
A	larger	faster
B	larger	slower
C	smaller	faster
D	smaller	slower

27. M/J/17/12/Q27

The graph shows how the height of a water wave varies with distance along the wave.



Which graph shows a wave with twice the amplitude, half the frequency, and the same speed?



28. O/N/16/11/Q21

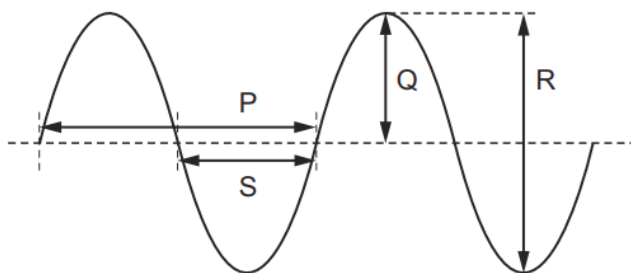
A phone rings and lights up.

Which types of wave are associated with the phone's operation?

	light	sound
A	longitudinal	longitudinal
B	longitudinal	transverse
C	transverse	longitudinal
D	transverse	transverse

29. O/N/16/11/Q22

The diagram shows the surface of water as a wave passes across a ripple tank.



Which lengths represent the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	Q	P
B	Q	S
C	R	P
D	R	S

30. O/N/16/12/Q25

Which statement about waves is correct?

- A** All electromagnetic waves are transverse waves.
- B** All transverse waves travel at 300 000 000 m/s.
- C** Longitudinal waves cannot be refracted.
- D** Water waves are longitudinal waves.

31. M/J/16/12/Q17

A water wave in a ripple tank refracts as it passes from deep water to shallow water.

Which properties change as the wave refracts?

- A frequency and amplitude
- B frequency and wavelength
- C speed and frequency
- D speed and wavelength

32. M/J/16/11/Q19

For a transverse wave, what is a *wavefront*?

- A a line joining all points on the same crest of a wave
- B a line showing the displacement of a wave
- C the energy content of a wave
- D the first part of a wave to reach a point

33. M/J/16/11/Q20

A longitudinal wave passes along a spring. The coils of the spring vibrate from side to side.

The diagram shows the positions of the coils at one particular time.

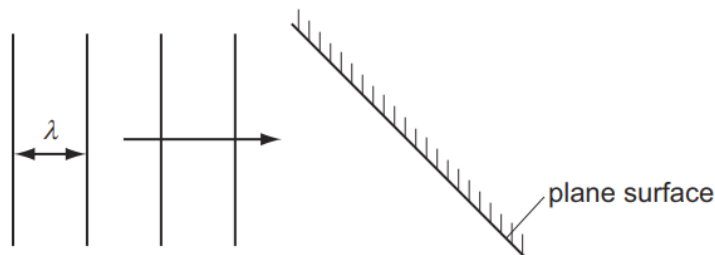


Which positions are one wavelength apart?

- A W and X
- B W and Z
- C X and Z
- D Y and Z

34. O/N/15/11/Q19

In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	larger	shorter
B	smaller	shorter
C	the same	longer
D	the same	the same

35. O/N/15/12/Q21

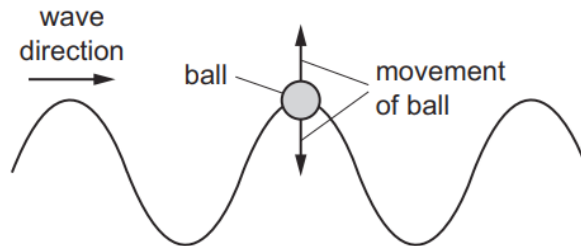
An echo sounder produces ultrasound of frequency 24 kHz. The ultrasound travels in water at a speed of 1.5 km/s.

What is the wavelength in water of ultrasound of this frequency?

- A** 0.063 m **B** 16 m **C** 36 m **D** 63 m

36. M/J/15/12/Q24

A ball floating in a ripple tank begins to move vertically up and down as a wave passes beneath it. The ball does not move horizontally.



Which statement is correct?

- A** Both energy and water are transferred in the wave direction.
B Energy is not transferred in the wave direction but water is.
C Energy is transferred in the wave direction but water is not.
D Neither energy nor water is transferred in the wave direction.

37. O/N/14/11/Q19

A wave of frequency 13 000 Hz travels 1300 m in 4.0 s.

What is the wavelength of the wave?

- A** 0.025 m **B** 0.40 m **C** 2.5 m **D** 40 m

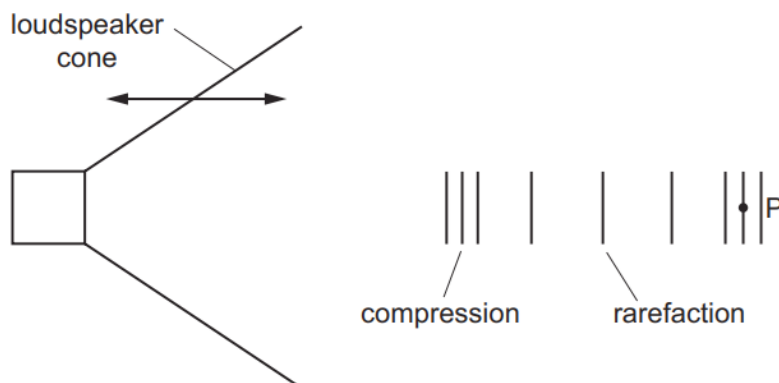
38. M/J/14/12/Q25

What is the frequency of a wave?

- A** The number of waves passing a fixed point per second.
B The number of peaks added to the number of troughs passing a fixed point per second.
C The time taken for one wave to pass a fixed point.
D The time taken for the displacement to change from maximum to minimum.

39. M/J/14/11/Q22

Compressions and rarefactions are sent out from a loudspeaker cone as it vibrates backwards and forwards. The frequency of vibration is 50 Hz.



A compression is at point P. How much time elapses before the next rarefaction arrives at P?

- A** 0.010 s **B** 0.020 s **C** 25 s **D** 50 s

40. M/J/14/11/Q23

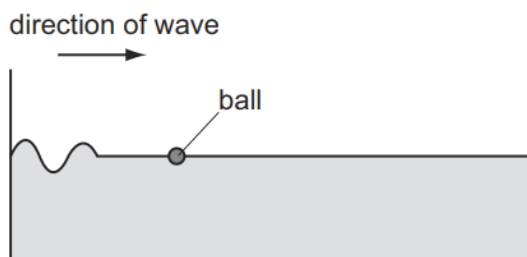
Water waves refract at a boundary between deep water and shallow water.

What is the effect on the frequency, wavelength and speed of the waves at the boundary?

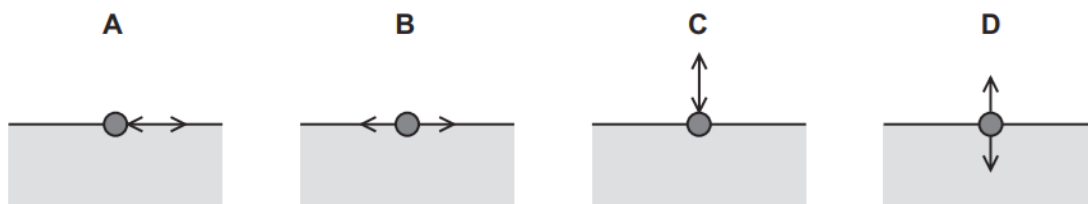
	frequency	wavelength	speed
A	changes	changes	stays the same
B	changes	stays the same	stays the same
C	stays the same	changes	changes
D	stays the same	stays the same	changes

41. O/N/13/11/Q19

The diagram shows a ball floating in a tank of water.

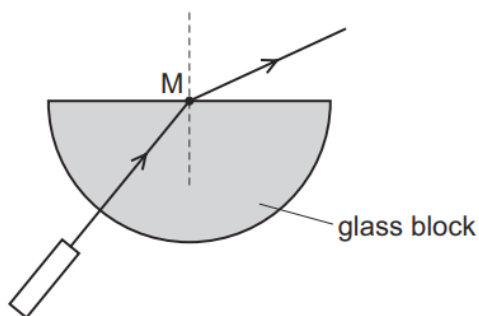


Which diagram shows the movement of the ball as the wave passes?



42. O/N/13/12/Q23

A ray of red light from a laser passes into a semi-circular glass block.



What is shown at M?

- A** dispersion
- B** rarefaction
- C** reflection
- D** refraction

43. M/J/13/12/Q22

Energy can be transferred in many different ways.

In which situation is energy transferred by wave motion?

- A** colliding atoms in a heated copper rod
- B** fast-moving electrons in a cathode-ray oscilloscope
- C** hot water rising in a heated saucepan
- D** ripples passing across water in a ripple tank

44. M/J/13/12/Q23

A wave has a frequency of 10 kHz.

Which pair of values of its speed and wavelength is possible?

	<u>speed</u> m/s	<u>wavelength</u> m
A	330	0.33
B	330	33
C	3.0×10^8	30
D	3.0×10^8	3.0×10^4

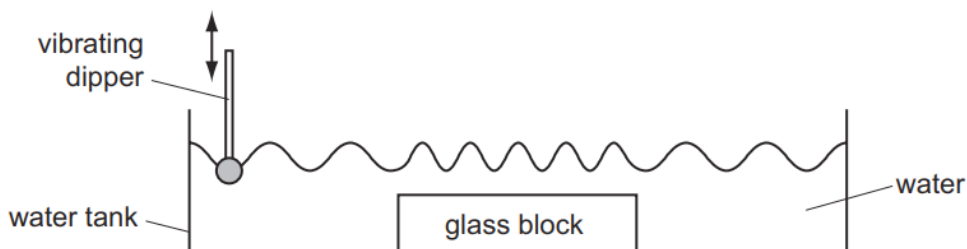
45. M/J/13/12/Q24

Which of the following travels as a longitudinal wave?

- A** a radio wave in air
- B** a sound wave in a solid
- C** a wave on a rope shaken from side to side
- D** an infra-red wave in space

46. M/J/13/11/Q17

A ripple tank is used to show wave behaviour. The dipper vibrates up and down at a constant frequency.

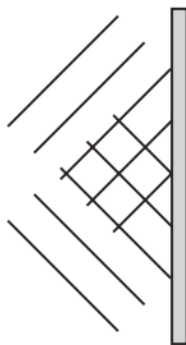


What happens to the frequency and to the speed of the wave as it reaches the glass block?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	remains the same	decreases
D	remains the same	increases

47. O/N/12/11/Q21

The diagram shows the pattern of waves in a ripple tank.

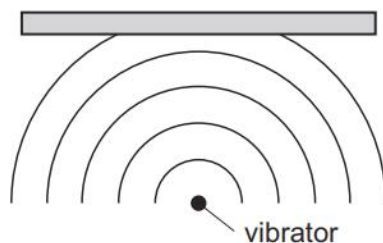


What does the pattern show?

- A** waves being reflected
- B** waves being refracted
- C** waves changing frequency
- D** waves changing speed

48. O/N/12/12/Q21

In a ripple tank, a vibrator produces circular wavefronts which hit a flat surface.



The reflected wavefronts are also parts of circles. Where is the centre of these circles?

A ●

B ●



D ● vibrator

49. M/J/12/12/Q21

A wave has a frequency of 2 Hz.

How many waves are produced in one minute?

A 2×60

B $\frac{60}{2}$

C 2

D $\frac{2}{60}$

50. M/J/12/11/Q21

The speed of radio waves is c . A radio station transmits waves with a wavelength λ .

What is the frequency of the transmission?

A $\frac{c}{\lambda}$

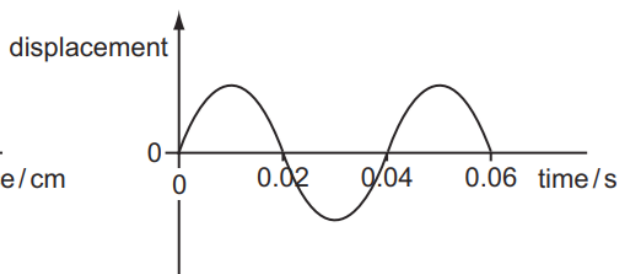
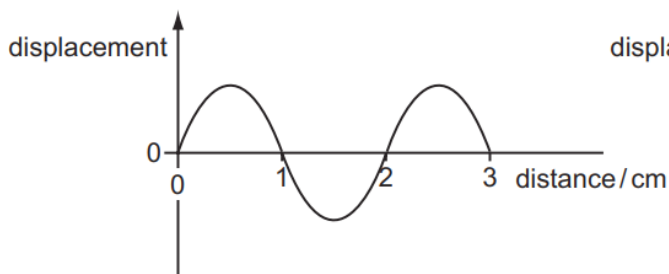
B $\frac{\lambda}{c}$

C $\frac{1}{\lambda}$

D $c\lambda$

51. M/J/11/12/Q22

The displacement-distance and displacement-time graphs are for a water wave in a ripple tank.



What is the speed of the water wave?

A 0.02 cm/s

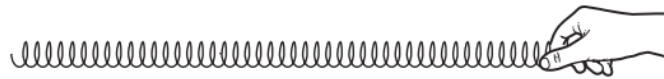
B 0.08 cm/s

C 25 cm/s

D 50 cm/s

52. O/N/11/11/Q23

A student uses a spring to demonstrate waves. He moves the spring with his hand.



spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?

A



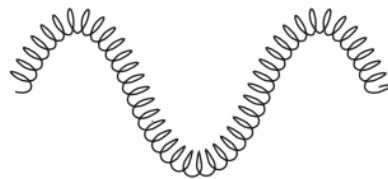
B



C



D



53. O/N/10/12/Q21

During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m/s.

What is the distance between the observer and the storm?

- A** 100 m **B** 200 m **C** 578 m **D** 1160 m

54. M/J/10/12/Q20

A flash of lightning and the corresponding sound of the thunder are detected 6 s apart. A student calculates that the lightning struck about 1800 m away.

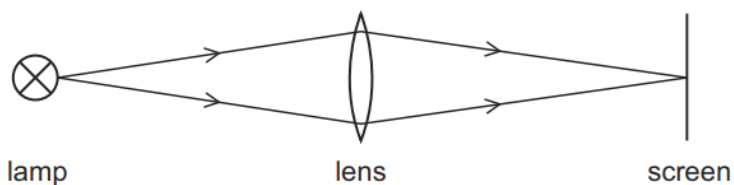
On which assumption is the calculation based?

- A** Light reaches us almost instantaneously, but sound travels at 300 m/s.
B Light travels 300 m/s faster than sound.
C Light travels 300 times faster than sound.
D The sound of the thunder was emitted 6 s after the flash.

55. M/J/10/12/Q19

Which diagram shows an example of a longitudinal wave?

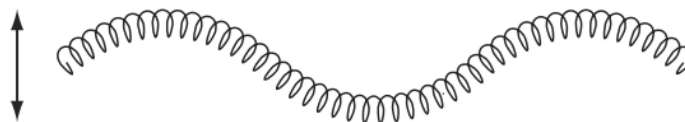
- A** light travelling from a lamp to a screen



- B** a spring pushed backwards and forwards



- C** a spring pushed up and down

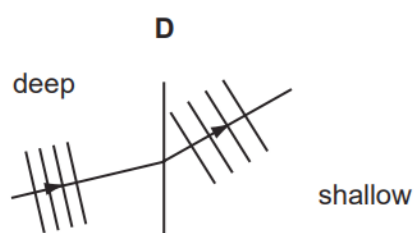
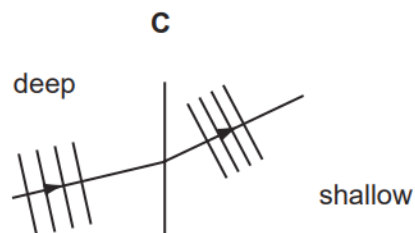
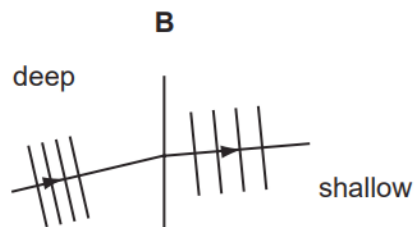
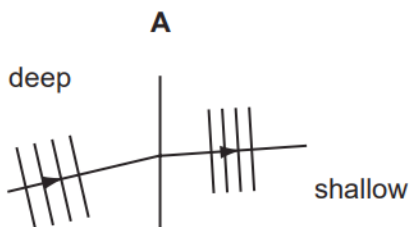


- D** a water ripple caused by a dipper moving up and down



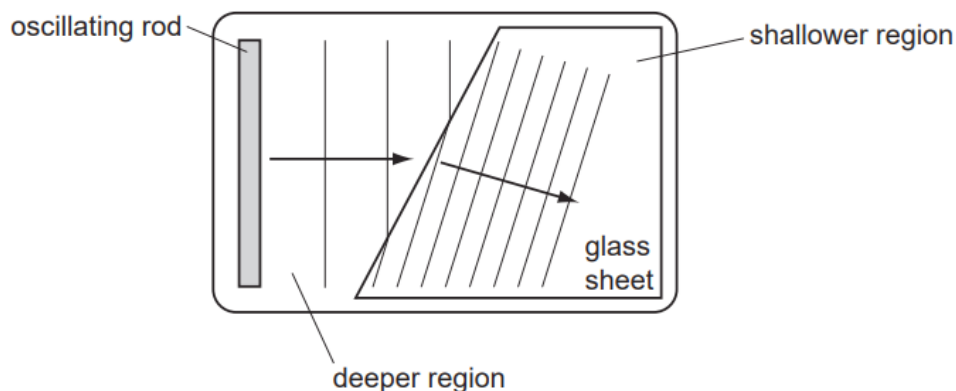
56. M/J/09/Q19

Which diagram correctly represents water waves travelling from deep water to shallow water?



57. O/N/08/Q22

The diagram shows the refraction of water waves in a ripple tank. The water is shallower above the glass sheet.

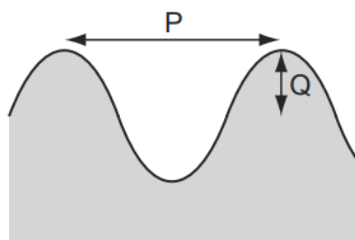


When crossing into the shallower region, what is the effect on the frequency and on the speed of the waves?

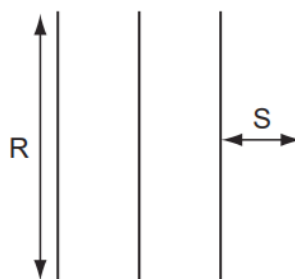
	wave frequency	wave speed
A	changes	changes
B	changes	unchanged
C	unchanged	changes
D	unchanged	unchanged

58. M/J/08/Q20

The diagrams show different views of a water wave in a ripple tank.



cross-section of wave



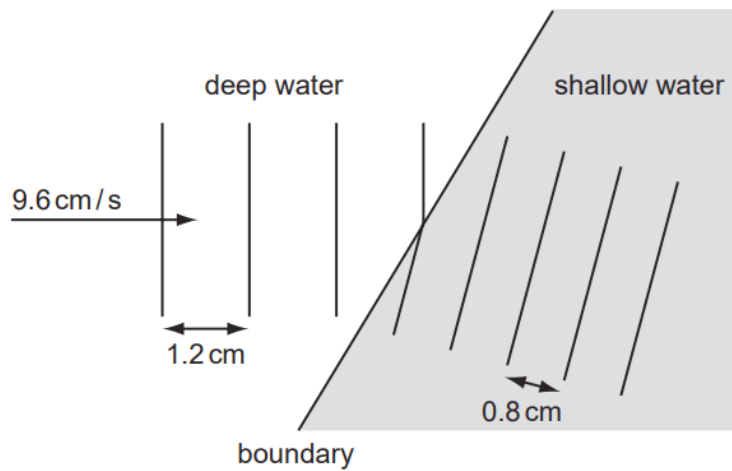
crests seen from above

Which letters represent a wavelength and a wavefront?

	wavelength	wavefront
A	P	R
B	P	S
C	Q	R
D	Q	S

59. M/J/08/Q21

A ripple tank is used to demonstrate refraction of plane water waves.



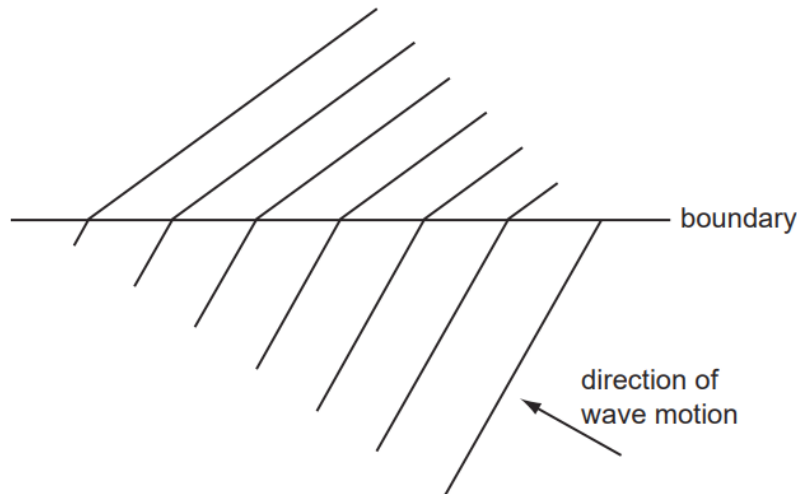
Waves in deep water have a wavelength of 1.2 cm and a speed of 9.6 cm/s. The wavelength of the waves in shallow water is 0.8 cm.

What is the speed of the waves in the shallow water?

- A** 6.4 cm/s **B** 8.0 cm/s **C** 9.6 cm/s **D** 14.4 cm/s

60. O/N/07/Q19

The diagram shows the refraction of water waves as they cross a boundary in a ripple tank.

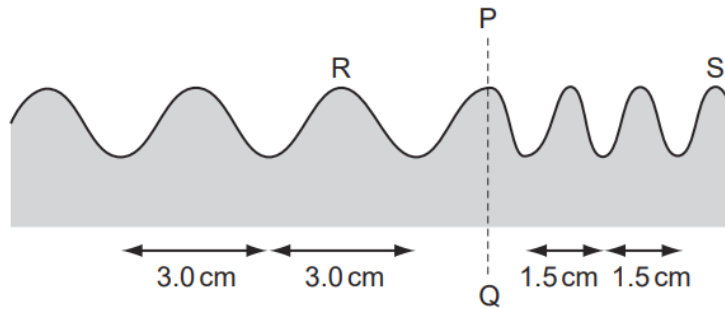


What causes this refraction?

- A** a change in frequency due to a change in depth
B a change in frequency due to a change in wavelength
C a change in speed due to a change in depth
D a change in speed due to a change in frequency

61. M/J/07/Q19

The diagram shows a water wave in a ripple tank.



The wave has a speed of 12 cm/s at R.

The wave crosses a boundary PQ where the distance between crests changes from 3.0 cm to 1.5 cm.

What is the speed of the wave at S?

- A** 3.0 cm/s **B** 6.0 cm/s **C** 12 cm/s **D** 24 cm/s

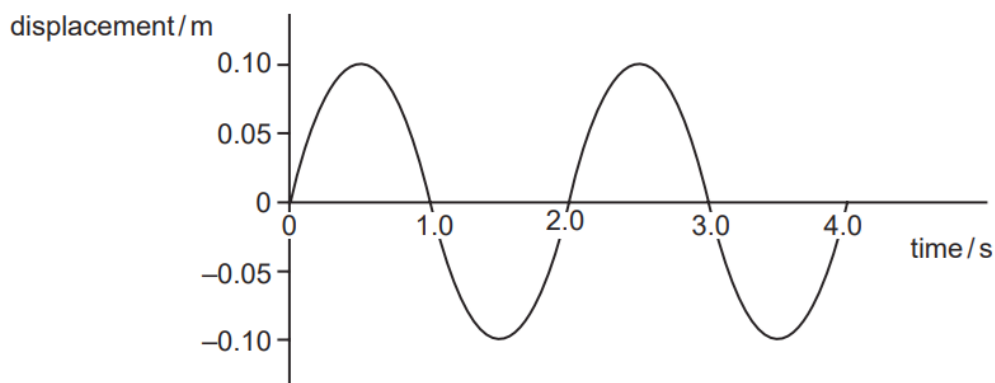
62. O/N/06/Q19

What is meant by the term *wavefront*?

- A** the distance between successive peaks of a wave
B the distance between the trough and the peak of a wave
C a line joining points along the peak of a wave
D a line joining the trough and the peak of a wave

63. O/N/06/Q20

The diagram shows how displacement varies with time as a wave passes a fixed point.

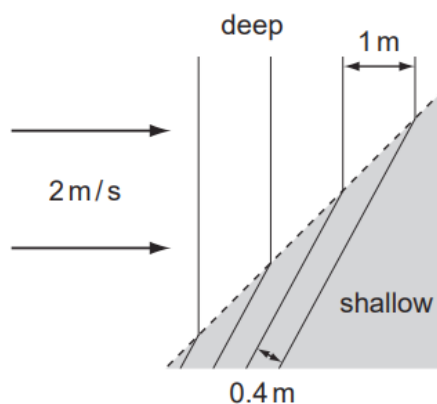


What is the frequency of this wave?

- A** 0.25 Hz **B** 0.50 Hz **C** 1.0 Hz **D** 2.0 Hz

64. M/J/06/Q22

Waves pass from deep water to shallow water and refraction occurs.

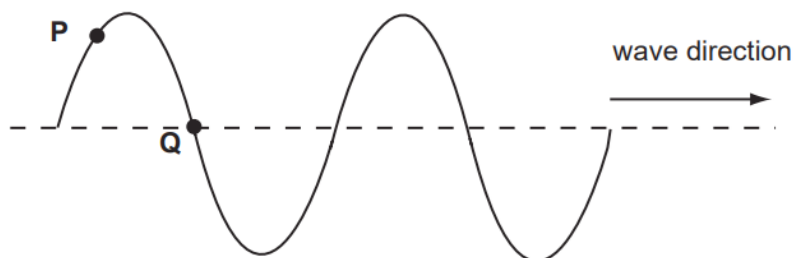


What is the speed of the waves in the shallow water?

- A** 0.2 m/s **B** 0.8 m/s **C** 2.0 m/s **D** 5.0 m/s

65. O/N/05/Q20

The diagram shows a wave on a string with two points **P** and **Q** marked. The wave is moving in the direction shown.



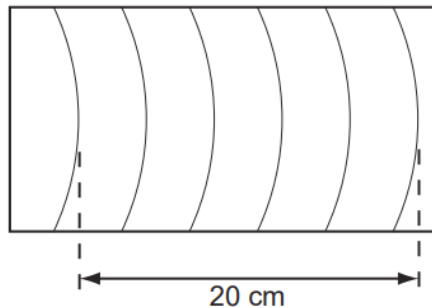
What will happen next?

- A** **P** will move to the right.
B **P** will move up.
C **Q** will not move.
D **Q** will move up.

66. O/N/05/Q21

The dipper in a ripple tank vibrates at a frequency of 4.0 Hz and the resulting wave pattern is photographed.

The distance between the two crests shown is 20 cm.

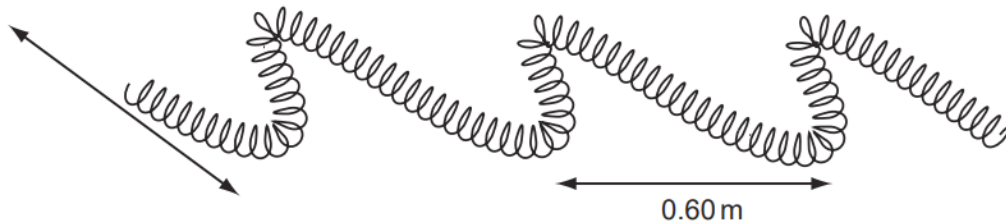


What is the speed of the wave?

- A** 4 cm/s **B** 5 cm/s **C** 16 cm/s **D** 20 cm/s

67. M/J/05/Q20

The diagram shows part of a spring that is shaken from side to side to produce a wave.



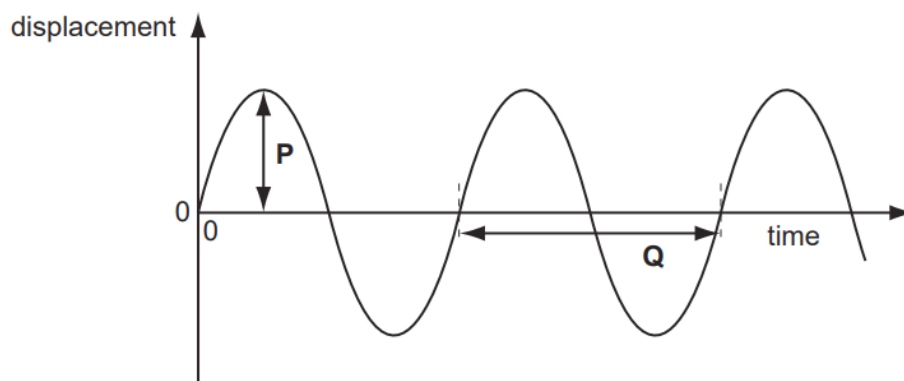
The distance between successive peaks is 0.60 m and the frequency is 2.5 Hz.

How long does it take for a wave to travel 3.0 m along the spring?

- A** 0.20 s **B** 0.50 s **C** 2.0 s **D** 5.0 s

68. O/N/04/Q22

The diagram shows a graph of wave motion.

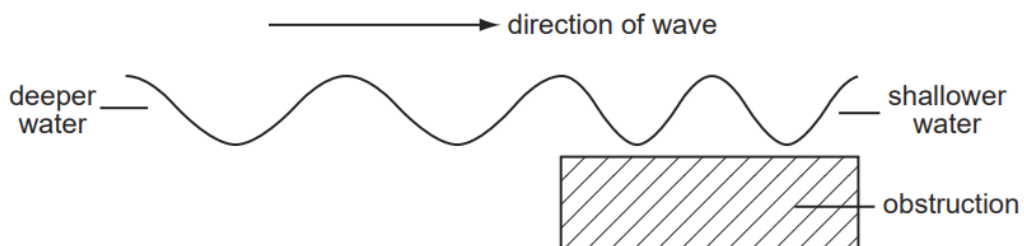


Which quantities are shown by distances **P** and **Q**?

	P	Q
A	amplitude	period
B	amplitude	wavelength
C	half the amplitude	period
D	half the amplitude	wavelength

69. M/J/04/Q19

The diagram shows a wave moving into shallower water.



The wavelength of the waves is reduced because

- A** both the frequency and the speed decrease.
- B** both the frequency and the speed increase.
- C** only the frequency increases.
- D** only the speed decreases.

ANSWERS

- | | |
|-------|-------|
| 1. A | 48. A |
| 2. A | 49. A |
| 3. D | 50. A |
| 4. D | 51. D |
| 5. B | 52. B |
| 6. A | 53. C |
| 7. B | 54. A |
| 8. B | 55. B |
| 9. B | 56. A |
| 10. A | 57. C |
| 11. D | 58. A |
| 12. C | 59. A |
| 13. B | 60. C |
| 14. A | 61. B |
| 15. A | 62. C |
| 16. D | 63. B |
| 17. A | 64. B |
| 18. C | 65. D |
| 19. B | 66. C |
| 20. C | 67. C |
| 21. C | 68. A |
| 22. B | 69. D |
| 23. D | |
| 24. A | |
| 25. A | |
| 26. A | |
| 27. D | |
| 28. C | |
| 29. A | |
| 30. A | |
| 31. D | |
| 32. A | |
| 33. D | |
| 34. D | |
| 35. A | |
| 36. C | |
| 37. A | |
| 38. A | |
| 39. A | |
| 40. C | |
| 41. D | |
| 42. D | |
| 43. D | |
| 44. D | |
| 45. B | |
| 46. C | |
| 47. A | |

1. M/J/23/22/Q5

Water waves are transverse waves.

- (a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

- (b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

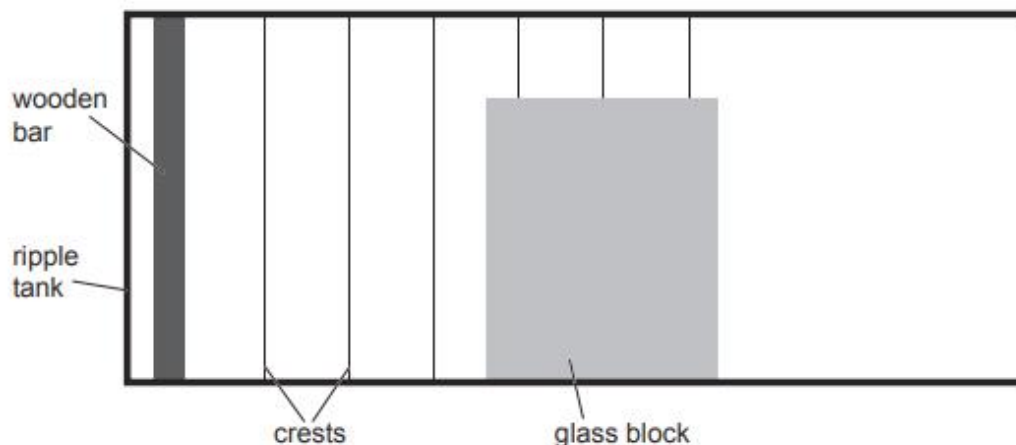


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

- (i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm

[3]

- (ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

- (iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
 [1]

- (iv) Describe how a decrease in wavelength affects the diffraction.

.....
 [1]

2. M/J/23/21/Q7

A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

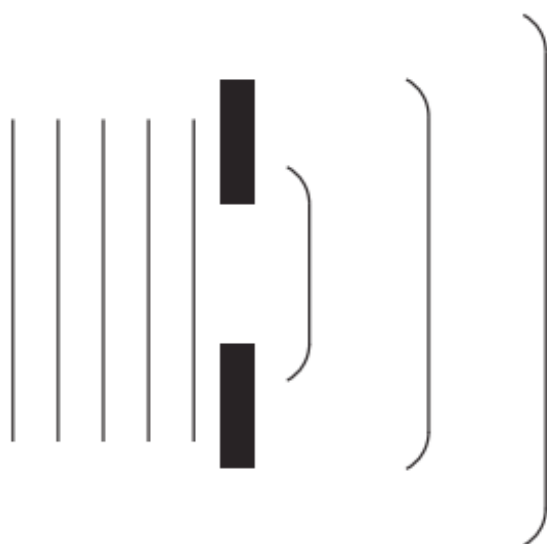


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
..... [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

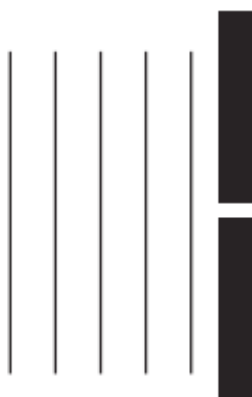


Fig. 7.2

[2]

- (c) In one part of the ripple tank, the water wave has a speed of 6.0 cm/s and a wavelength of 2.0 cm.

The wave then passes into a shallower region of the tank.

The speed of the wave in the shallow region is 4.0 cm/s.

- (i) Define the term 'wavelength'.

.....
..... [1]

- (ii) Calculate the frequency of the wave.

frequency = Hz [2]

- (iii) Calculate the wavelength of the wave in the shallow part of the tank.

wavelength = cm [1]

[Total: 7]

3. M/J/22/22/Q8

(a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

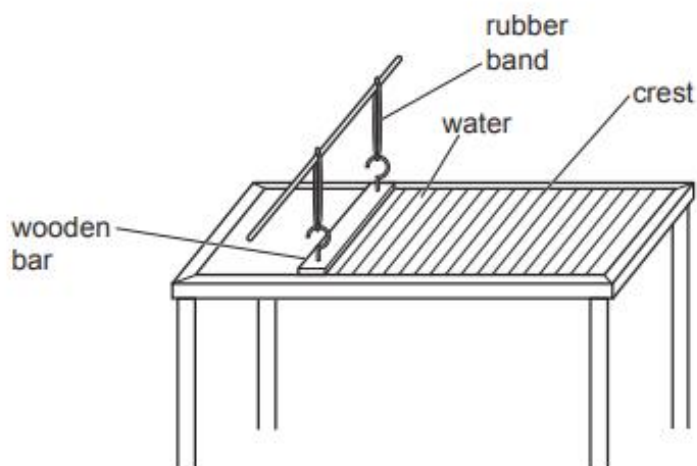


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(i) Calculate the number of crests produced in 1.5 s.

number = [1]

(ii) The height of the wave is measured from the level of the undisturbed surface of the water. The height of the wave at one point is 0 at time = 0.

On Fig. 8.2, draw a graph to show how the height of the wave at this point varies with time.

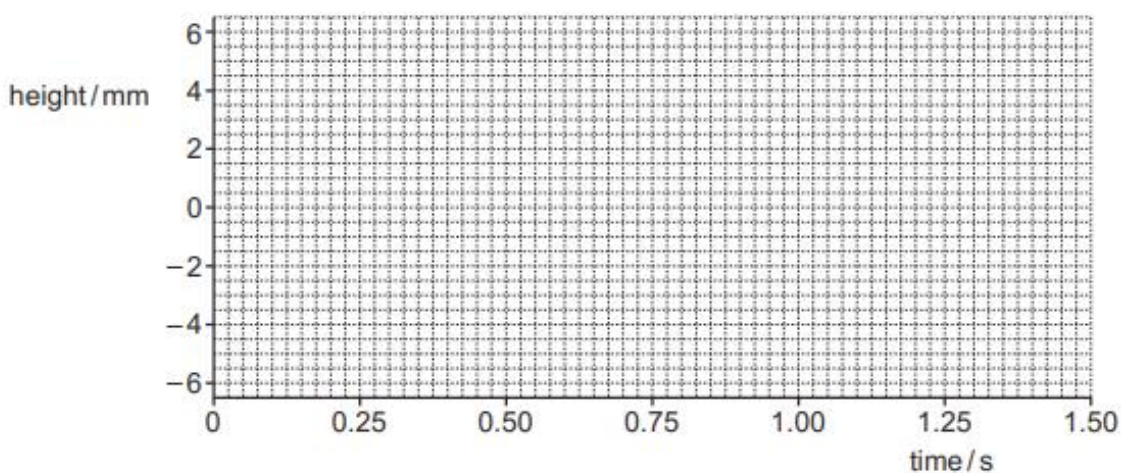


Fig. 8.2

[2]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
..... [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed

wavelength [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.



Fig. 8.3

[3]

4. M/J/21/22/Q10

- (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

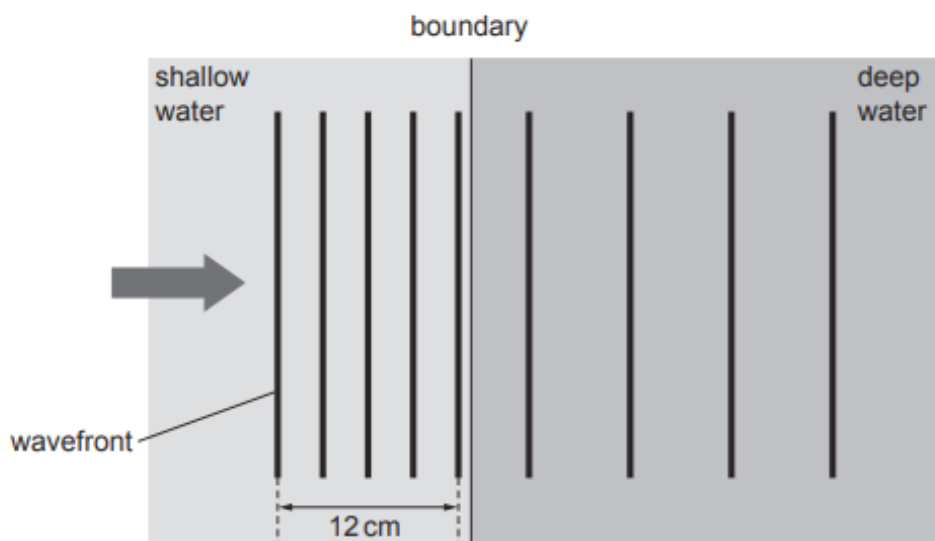


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (i) Determine the wavelength of the wave in the shallow water.

wavelength = [1]

- (ii) Calculate the frequency of the wave produced.

frequency = [2]

- (iii) Determine the speed of the water wave in the shallow water.

speed = [2]

- (iv) On Fig. 10.2, draw lines to show what happens to the wavelength, the frequency and the speed of the water wave as it enters the deep water.

One line has been drawn for you.

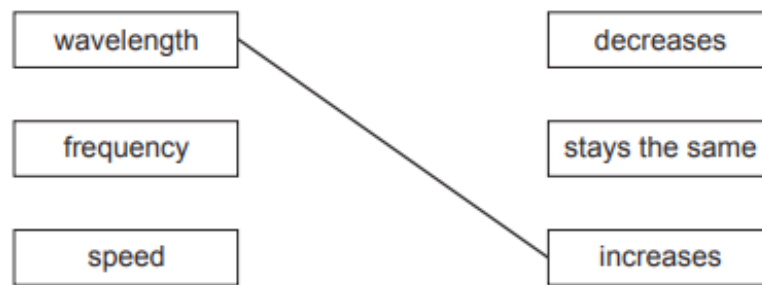


Fig. 10.2

[1]

- (v) The direction of the boundary is now altered. The water wave now enters the deep water from the shallow water at an angle, as shown in Fig. 10.3.

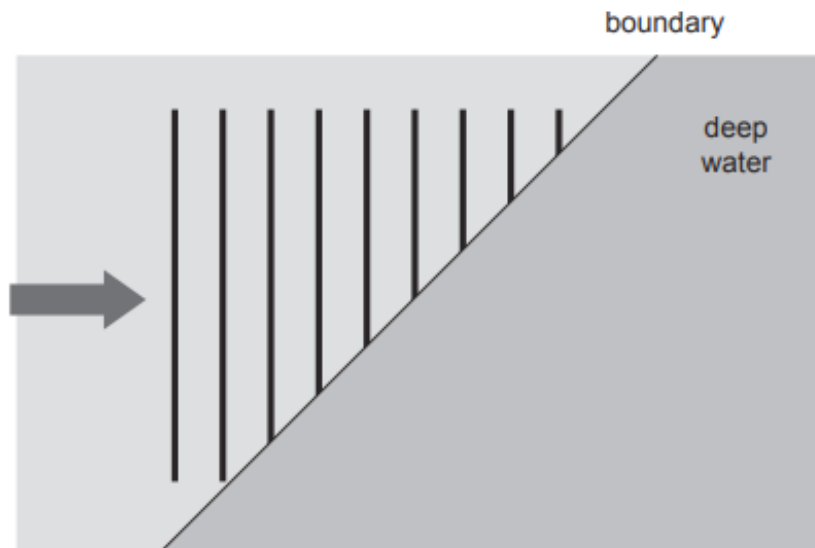


Fig. 10.3

On Fig. 10.3, draw the wavefronts in the deep water.

[3]

5. M/J/19/21/Q10

In Fig. 10.1 the vertical lines represent the crests of a wave in a ripple tank.

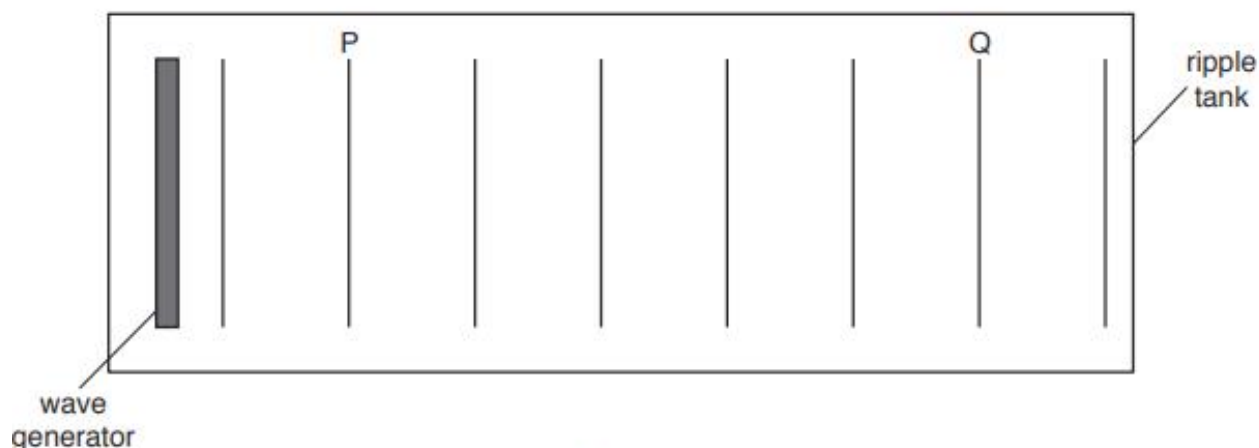


Fig. 10.1

Fig. 10.1 is drawn to scale, with 1.0 cm on the diagram representing 4.0 cm in the tank.

A crest takes 2.0 s to travel from P to Q.

(a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

(ii) Define the term *frequency*.

.....
..... [2]

(iii) Determine the frequency of the wave.

frequency = [1]

- (iv) The wave generator is a bar of wood.

Describe how it is used to produce the wave shown.

.....
.....
..... [2]

- (v) Some water in the ripple tank is removed. The water becomes less deep. No other change is made.

State how the change in depth affects the wavelength and the speed of the wave.

speed

wavelength

[1]

6. O/N/18/21/Q9

Some waves are longitudinal and some waves are transverse.

- (a) State how a longitudinal wave differs from a transverse wave.

.....
 [1]

- (b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

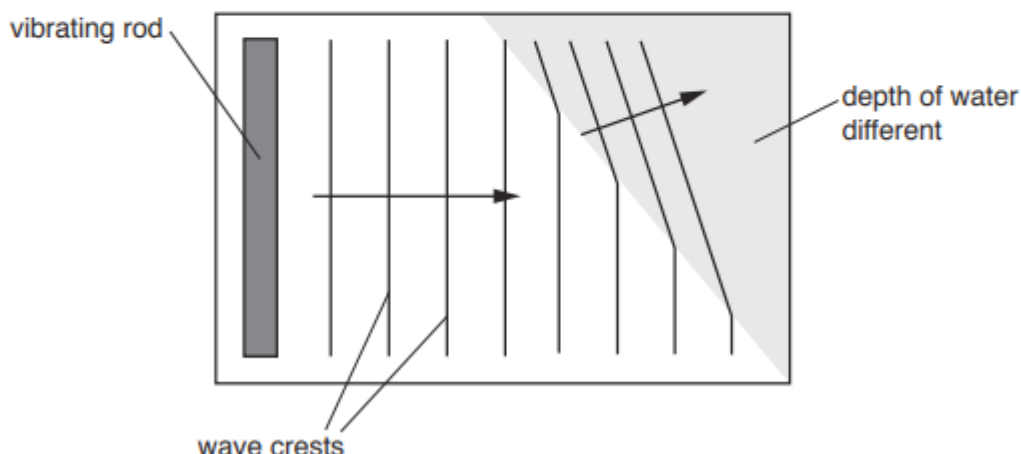


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

- (i) In the left-hand section of the tank, the wavelength of the wave is 0.019m and it travels at 0.17m/s.

Calculate the frequency of the wave.

frequency = [2]

- (ii) State what happens to the frequency of the wave, as it passes into the right-hand section of the tank.

..... [1]

- (iii) Using Fig. 9.1, state and explain what happens to the speed of the wave as it passes into the right-hand section of the tank.

.....

 [2]

7. M/J/18/21/Q11

A water wave is produced in a ripple tank.

- (a) (i)** Draw a diagram of a ripple tank and describe how it is used to produce and observe the wave.

.....

.....

.....

.....

.....

.....

.....

.....[4]

- (ii)** The water wave in the ripple tank transfers energy.

Describe how to show that there is a transfer of energy.

.....

.....

.....

.....[2]

- (iii)** State what is meant by a *wavefront*.

.....

.....

.....[2]

- (b) Fig. 11.1 shows the variation with time t of the height of water at the same place in the tank for two waves P and Q. The height is sometimes above and sometimes below the mean level which is marked as 0.

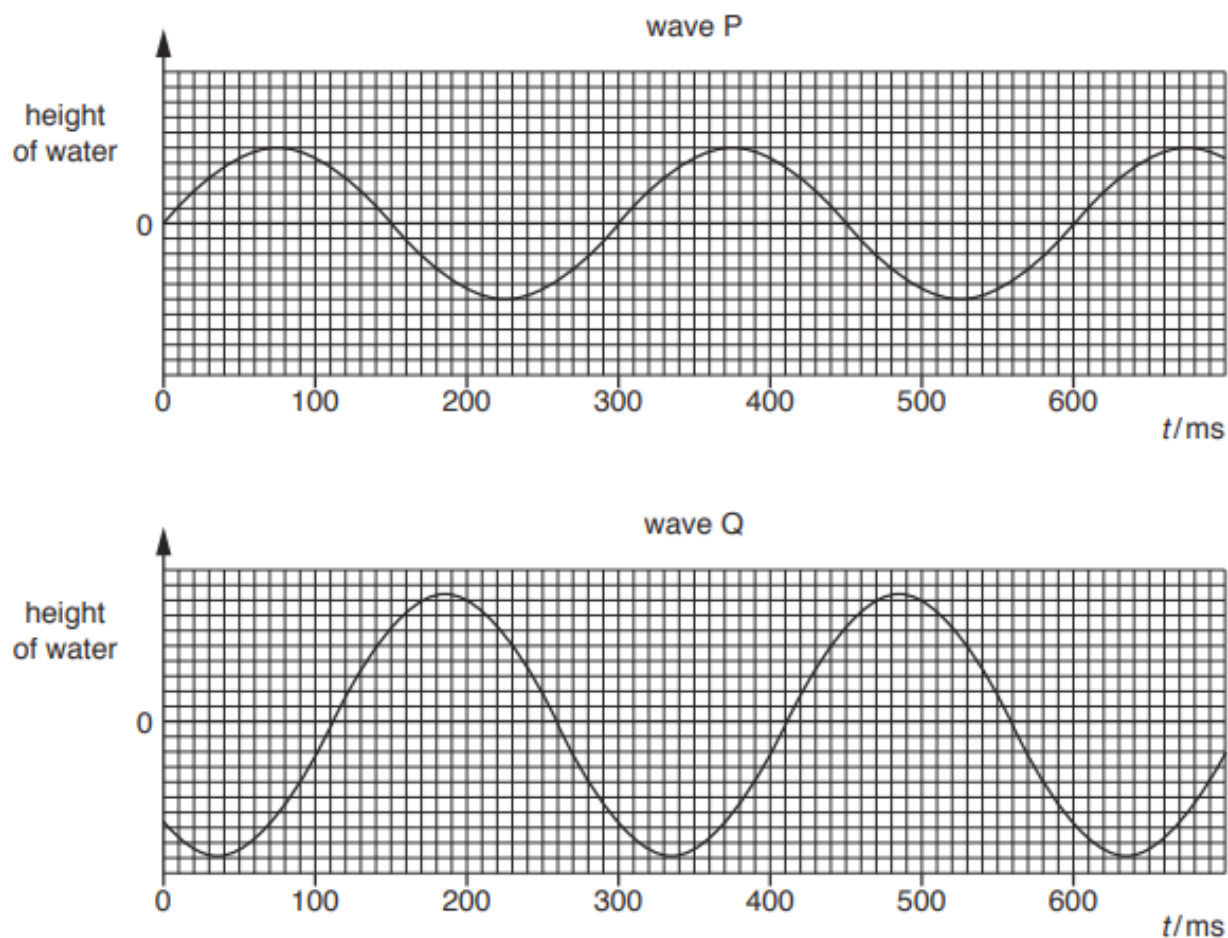


Fig. 11.1

- (i) State one difference and one similarity of the two waves.

difference

.....

similarity

.....

[2]

(ii) The speed of the water wave is 0.20 m/s.

Determine

1. the frequency of wave P,

frequency =[3]

2. the wavelength of wave P.

wavelength =[2]

8. M/J/17/22/Q6

- (a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b)** A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i)** Calculate the wavelength of the sound.

wavelength =[2]

- (ii)** A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

9. M/J/16/21/Q5

- (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.

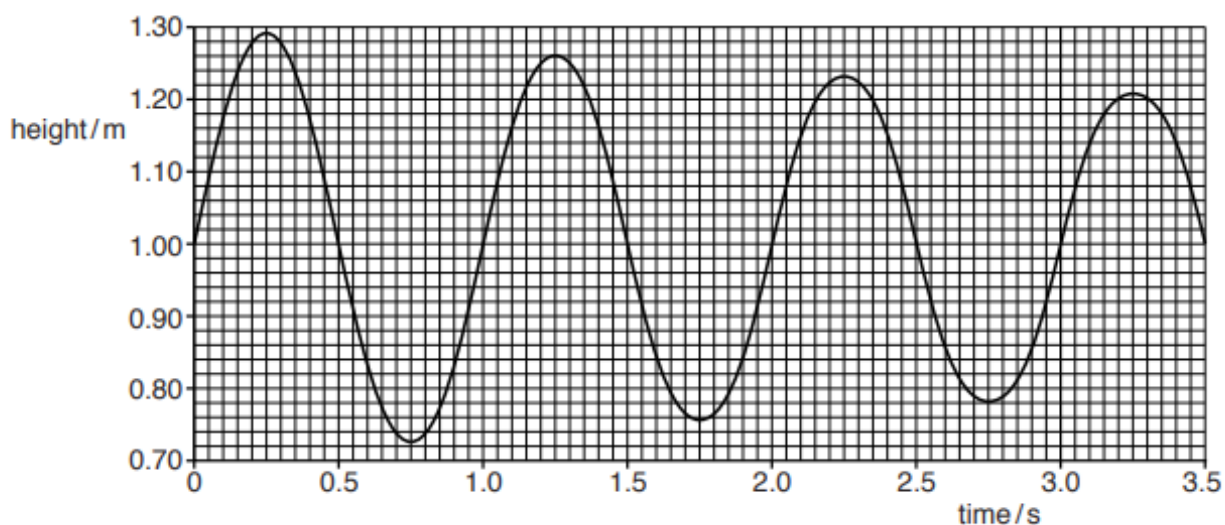


Fig. 5.2

- (i) Determine the amplitude of the vibration at time = 0.25 s.

.....[1]

- (ii) Describe how the graph shows that the frequency of the wave is constant.

.....
[1]

10. M/J/15/22/Q9

Fig. 9.1 shows the wavefronts of a water wave in deep water in a ripple tank.

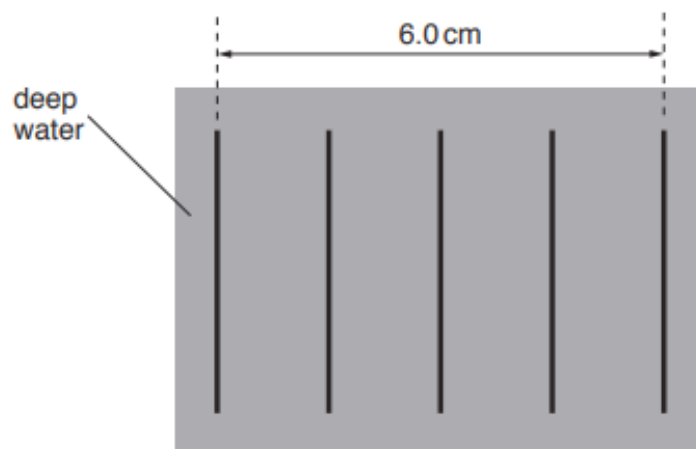


Fig. 9.1

The frequency of the wave in deep water is 5.0 Hz.

(a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

(b) (i) Determine the wavelength of the wave in deep water.

..... [1]

(ii) Calculate the speed of the wave in deep water.

speed = [2]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

.....

 [2]

- (ii) The wave in deep water shown in Fig. 9.2 travels towards the right and enters the shallow water at an angle. The wave refracts.

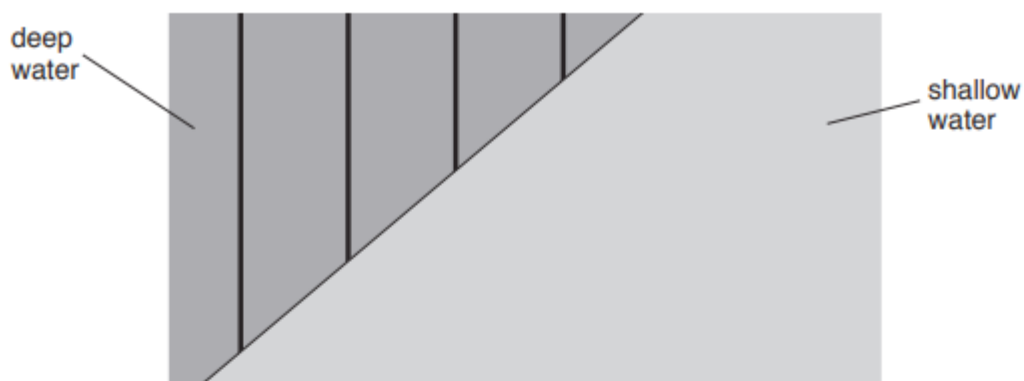


Fig. 9.2

On Fig. 9.2, draw the wavefronts in the shallow water.

[3]

- (d) Sound is also a wave.

- (i) Describe one difference between a sound wave and a water wave.

.....

.....

.....

..... [2]

- (ii) The speed of sound in carbon dioxide gas is less than the speed of sound in air. Using this information, or otherwise, describe an experiment to show the refraction of sound waves. You may include a diagram of your apparatus.

.....

.....

.....

.....

.....

..... [3]

11. M/J/15/21/Q4

Fig. 4.1 shows a wave on a string. The wave is travelling towards the right.

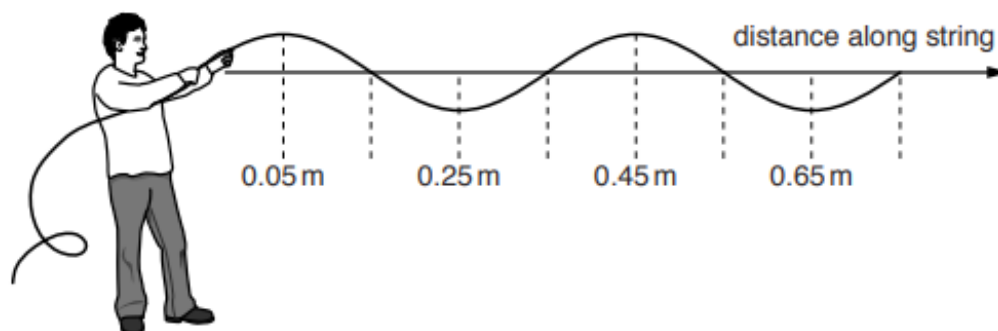


Fig. 4.1

- (a)** Describe the movement of the particles of the string.

.....

.....

.....

.....[2]

- (b)** Determine the wavelength of the wave.

wavelength =[1]

- (c)** The speed of the wave is 2.0 m/s.

- (i)** Calculate the frequency of the wave.

frequency =[2]

- (ii)** On Fig. 4.1, draw the string at a time 0.10 s later than the time in Fig. 4.1. [1]

12. O/N/14/21/Q4

Ripple tanks are often used to illustrate wave motion.

- (a) Describe what is meant by *wave motion* in a ripple tank.

.....
.....
.....[2]

- (b) When describing wave motion, state what is meant by

- (i) *frequency*,

.....[1]

- (ii) *wavelength*.

.....
.....[2]

- (c) A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.

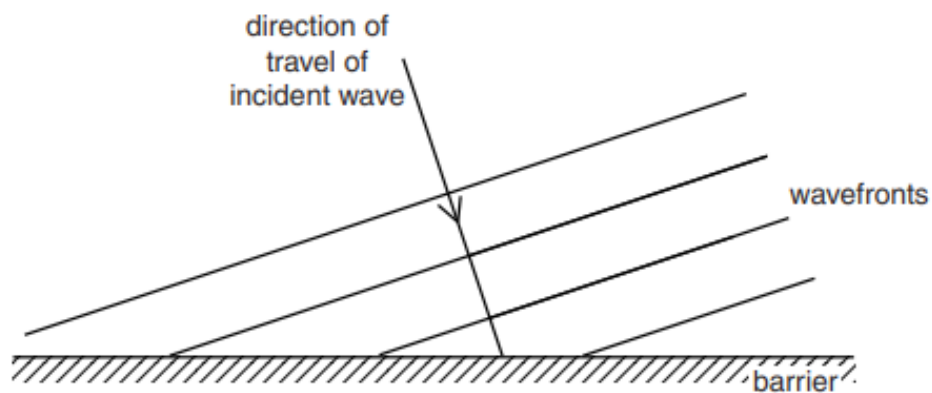


Fig. 4.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 4.1, draw the reflected parts of these three wavefronts.

[3]

13. M/J/14/22/Q4

Fig. 4.1 shows part of a long rope used by a student to show a transverse wave.



Fig. 4.1

(a) On Fig. 4.1,

(i) mark the direction of movement of the student's hand, [1]

(ii) mark and label the wavelength λ of the wave, [1]

(iii) mark and label the amplitude A of the wave. [1]

(b) Describe how the frequency of the wave is found using a stopwatch.

.....
.....
.....
..... [2]

(c) Using the same rope, the student produces a wave of a longer wavelength than that shown in Fig. 4.1.

State how the student does this.

.....
..... [1]

14. O/N/12/21/Q5

A water wave of frequency 3.6Hz is generated in a ripple tank. Fig. 5.1 is a scale diagram that shows the wavefronts travelling from deep water to shallow water in the ripple tank. A length of 1.0 cm on the diagram represents 8.0 cm in the ripple tank.

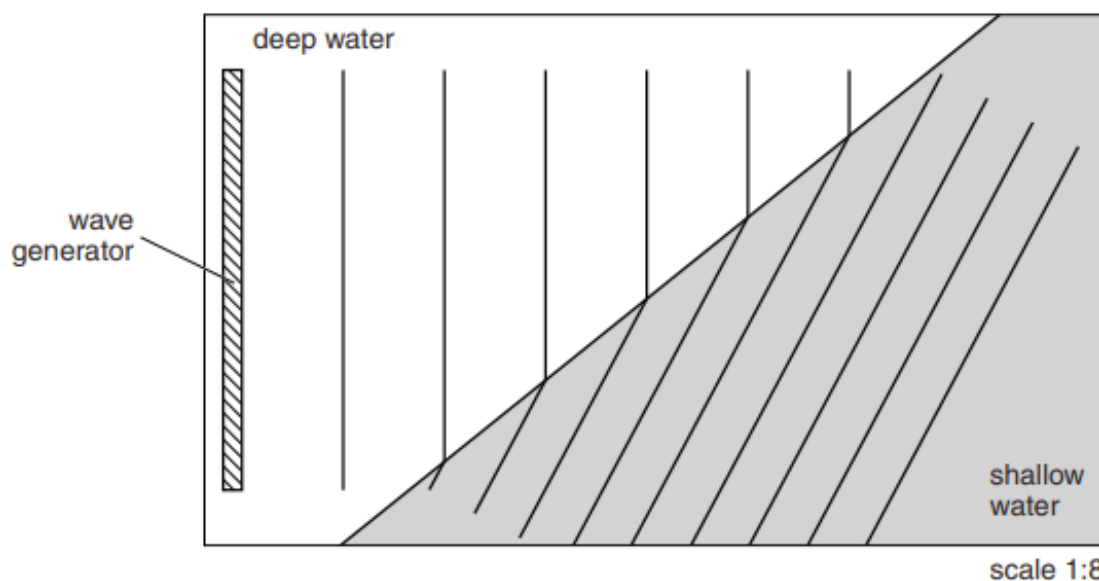


Fig. 5.1

- (a) (i) Using measurements from Fig. 5.1, determine the wavelength of the wave in the deep water.

wavelength = [2]

- (ii) Calculate the speed of the wave in the deep water.

speed = [2]

- (b) As the wave passes into the shallow water, state what happens to

- (i) the frequency,

..... [1]

- (ii) the speed.

..... [1]

15. M/J/12/22/Q6

Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.



Fig. 6.1 (not to scale)

(a) Describe what is meant by *wave motion*.

.....

.....

.....[2]

(b) On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X.
[1]

(c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

(i) Calculate the speed of the wave.

speed =[2]

(ii) On Fig. 6.1, mark a distance which shows how far a wavefront at X moves in 1.0 s.
Label this distance *D*. [1]

16. M/J/11/22/Q4

Fig. 4.1 shows circular wavefronts produced at the centre of a circular ripple tank.

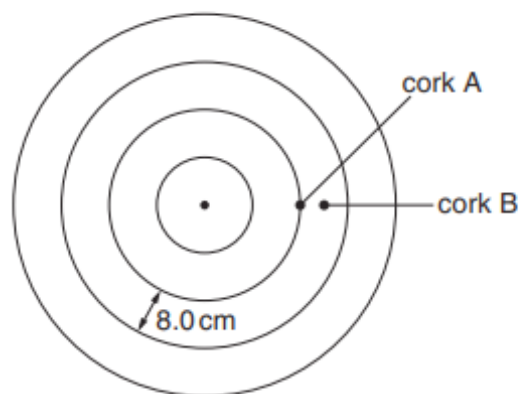


Fig. 4.1

Two corks, A and B, float on the water in the ripple tank. They move up and down on the surface of the water as the wave passes. The wavelength of the wave is 8.0 cm.

Fig. 4.2 shows how the displacement of A varies with time.

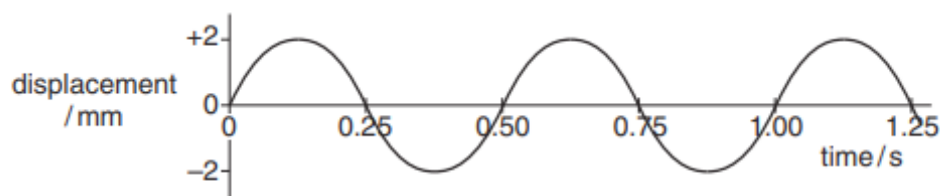


Fig. 4.2

- (a) State the amplitude of the vibrations of A as the wave passes.

amplitude =[1]

- (b) The horizontal distance between A and B is half the wavelength of the wave.

On Fig. 4.2, sketch a graph to show how the displacement of B varies with time. [2]

- (c) (i) Use Fig. 4.2 to determine the frequency of the wave.

frequency =[2]

- (ii) The distance between the centre of the ripple tank and its edge is 40 cm.

Determine the time taken by a wavefront to travel from the centre of the tank to the edge.

time =[2]

17. M/J/10/21/Q4

Fig. 4.1 and Fig.4.2 are diagrams of a ripple tank being used to show two properties of waves.

(a) Fig. 4.1 shows wavefronts approaching a barrier in the water.

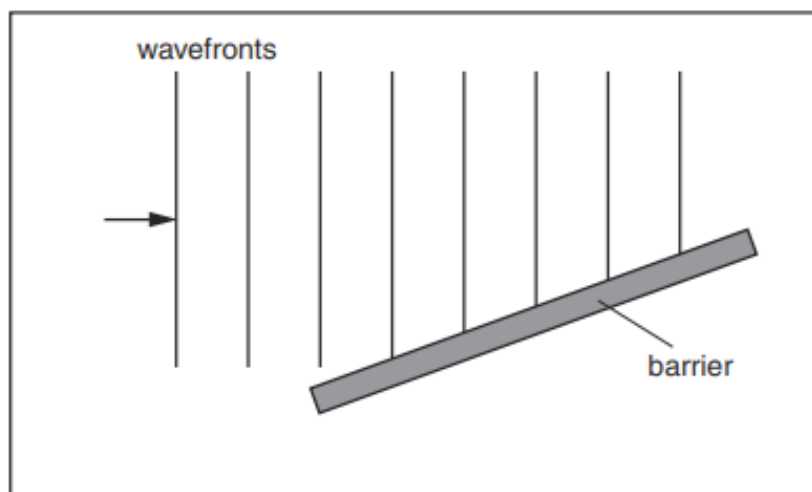


Fig. 4.1

Complete Fig. 4.1 to show the reflection of the wavefronts at the barrier.

[2]

(b) Fig. 4.2 shows the wavefronts approaching shallow water above a piece of glass.

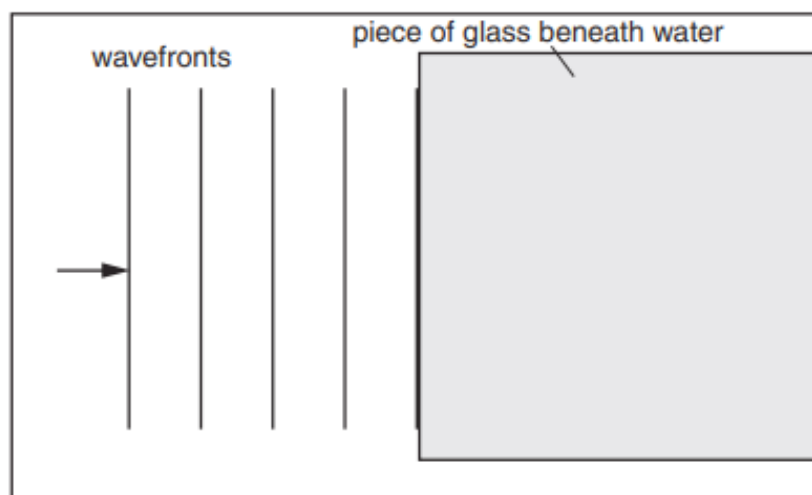


Fig. 4.2

(i) Complete Fig. 4.2 to show the wavefronts in the shallow water.

[1]

(ii) As the wave passes into the shallow water, state what, if anything, happens to

1. the wave speed,[1]

2. the frequency.[1]

18. O/N/05/Q3

A student produces wavefronts in a ripple tank to demonstrate refraction, as shown in Fig. 3.1. He places a sheet of glass under the water on the right-hand side of the tank. The arrows show the directions of movement of the wavefronts.

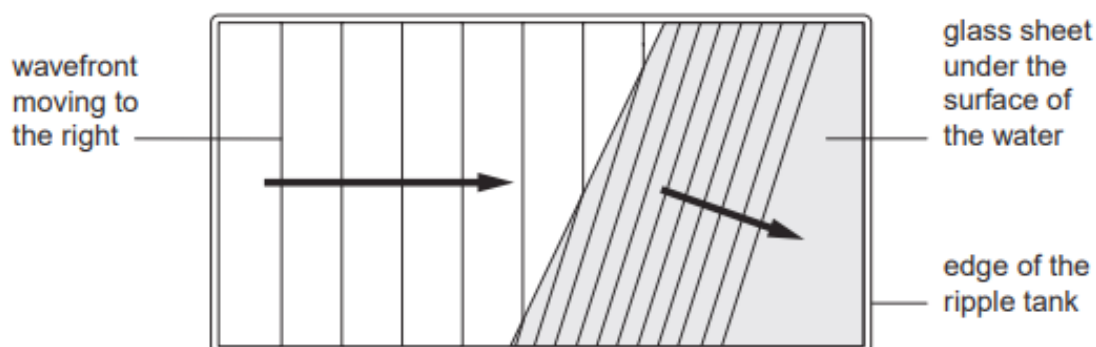


Fig. 3.1

- (a) State what is meant by a *wavefront*.

.....

 [1]

- (b) State what happens to each of the following quantities as the wavefronts change direction.

(i) wavelength [1]

(ii) speed [1]

(iii) frequency [1]

19. O/N/04/Q5

Fig. 5.1 shows part of a long, thin spring used to demonstrate a transverse wave.

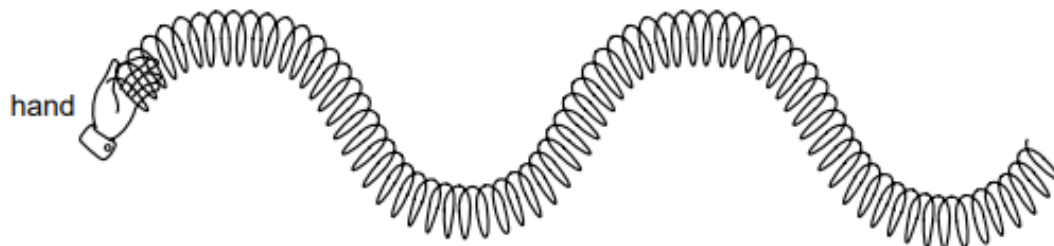


Fig. 5.1

The wave shown in Fig. 5.1 has a frequency of 4.0 Hz.

(a) (i) On Fig. 5.1, mark the direction the hand must move to make a transverse wave.

(ii) Describe how the hand must move to make a transverse wave of frequency 4.0 Hz.

.....

.....

.....

[2]

(b) The speed of the wave is 0.80 m/s. Calculate its wavelength.
State clearly the formula that you use.

wavelength = [3]

(c) State what must be done to double the wavelength of the wave on the spring.

.....

.....

.....[1]

ANSWERS

1.

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1
5(b)(iv)	less diffraction or less bending	B1

2.

Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz)	A1
7(c)(iii)	1.3 (cm)	B1

3.

Question	Answer	Marks
8(a)(i)	3	B1
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	B1
	1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i)	B1
8(b)(i)	move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight	B1
8(b)(ii)	<i>speed</i> remains the same	B1
	<i>wavelength</i> decreases	B1
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary	B1
8(c)(ii)	diagram showing within the ripple tank:	B1
	crests parallel to bar hitting boundary with a boundary labelled	
	correct refraction of plane crests either towards or away from normal on the same side of the normal or no direction change if crests are initially parallel to boundary but with a wavelength change	B1
	different wavelength shown before and after refraction and constant before and after (by eye)	B1

4.

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second or $5 / 10$ or $f = 1 / T$	C1
	2(.0) Hz	A1
10(a)(iii)	$(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same and speed increases	B1
10(a)(v)	$\geq$ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1

5.

Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	C1
	6.4–7.2 cm	A1
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	C1
	number of wavelengths/crests/troughs/ <u>wave</u> cycles/ <u>wave</u> oscillations generated/made/pass a point in one second/unit time	A1
	or number of oscillations of a <u>source/particle</u> in one second / unit time	
10(a)(iii)	2.5 Hz	B1
10(a)(iv)	moved up and down	B1
	regularly or at constant frequency / 2.5 times a second	B1
10(a)(v)	<i>speed</i> smaller / decreases and <i>wavelength</i> smaller / decreases	B1

6.

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or $0.17 / 0.019$	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1

7.

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	<i>difference</i> : Q has a greater / different amplitude or phase	B1
	<i>similarity</i> : same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	(f=) $1/t$ numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(iii)	$2(\lambda =) v/f$ or $v = \lambda f$ – numerical or algebraic in any form	C1
	0.060 m	A1

8.

Question	Answer	Marks
6(a)	longitudinal – vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v/f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

9.

- (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
- water*: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time/period for one wave(length)/cycle constant B1
or each oscillation/cycle takes one second

10.

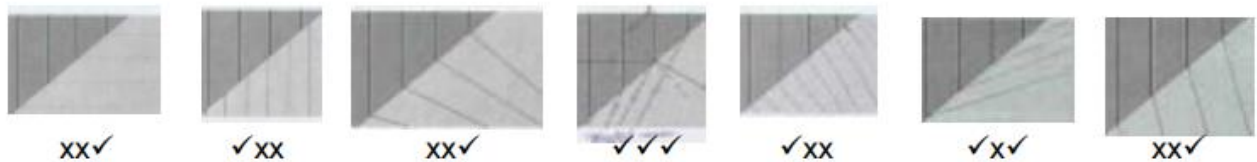
- (a) number of waves (that pass a point) M1
 or number of oscillations (passing a point)
 in unit time or per second or in 1 second A1

- (b) (i) 1.5 cm B1

- (ii) $(v =)f\lambda$ or 5×1.5 seen C1
 or $(s=)d/t$ and $f = 1/t$
 7.5 cm/s A1

- (c) (i) wavelength decreases B1
 travels a shorter distance in the same time B1
 or frequency stays the same (and $v = f\lambda$)

- (ii) wavefronts with smaller wavelength (by eye) B1
 smaller angle to surface (by eye) and slanted down B1
 wavefronts join those in shallow water B1



- (d) (i)

sound	water
particles / wave / source - vibrate / oscillate / move in direction of (travel of) wave / along wave - move backwards and forwards	particles / wave / source - vibrate / oscillate / move at 90° to direction of (travel of) wave - move up and down
(contains) compressions and rarefactions or particles come closer / further apart	(contains) crests and troughs
speed 300–330 m/s	wave slower (than sound)

B1
 B1
 one row only

- (ii) method of generating sound, e.g. (loud) speaker (and signal generator) B1
 apparatus that enables refraction clear, e.g. carbon dioxide in balloon or any shape where refraction is possible B1
 method of detecting refraction, e.g. microphone and how it is used to show refraction B1

11.		
(a)	oscillate / vibrate stated or described transverse movement described	B1 B1
(b)	0.40 m	B1
(c) (i)	$v = f \lambda$ or $(f =) v / \lambda$ or $2 / (b)$ 5.0 Hz	C1 A1
(ii)	clear attempt to draw wave moved along 0.20 m to right	B1
12.		
(a)	any two from: transmission of energy without net movement of medium through vibration of particles	B2
(b) (i)	number of (complete) waves / cycles / oscillations per unit time / second	B1
(ii)	distance between (neighbouring) waves distance between (neighbouring) wavefronts / points of same phase or crest to crest / trough to trough distance	C1 A1
(c)	three reflected wavefronts roughly correct direction wavelengths equal to each other and incident wavelength by eye reflected wavefronts joined to incident wavefronts	M1 A1 B1
13.		
(a) (i)	up and down clear, e.g. by double headed arrow or down	[B1]
(ii)	any correct distance between consecutive points in phase	[B1]
(iii)	correct distance	[B1]
(b)	measure number of oscillations / count waves (passing) in a stated time or time at least one oscillation show how to calculate number of oscillations per second	[B1] [B1]
(c)	moves (hand or rope) with slow(er) speed or rate / less frequency / less times per sec	[B1]

14.

- (a) (i) clear attempt at measuring more than one wavelength e.g. $6.85/5$
or $1.30 - 1.45 \text{ cm}$ B1
 $10.7 - 11.3 \text{ cm}$ B1
- (ii) $(v =)f\lambda$ or $3.6 \times (a)(i)$ C1
 $40(39.6) \text{ cm/s}$ A1
- (b) (i) stays the same B1
- (ii) decreases B1

15.

- (a) Any 2 of
 - an oscillation/vibration/movement up and down
 - carries energy
 - no (net) movement of the medium/transfer of matter
B2
- (b) arrow downwards or upwards or both B1
- (c) (i) $(v =)f\lambda$ in any form numerical or algebraic C1
 $5(.0) \text{ cm/s}$ or $0.05(0) \text{ m/s}$ A1
- (ii) line or indication **labelled D** of length 2 wavelengths B1

16.

- (a) $2(.0) \text{ mm}$ B1
- (b) same period (by eye), with at least one wave
opposite phase to wave drawn B1
B1
- (c) (i) $(f =)1/T$ numerical or algebraic seen (e.g. $1/0.5$) // 1 wave in 0.5 s // 2 waves
in 1 s C1
 $2(.0) \text{ Hz}$ A1
- (ii) $v = f\lambda$ // 8×2 or $8 \times (i)$ // 16 (cm/s) // 5 (wavelengths from centre to edge) //
 $(t =)d/v$ C1
 2.5 s **ecf** from (i) – i.e. **accept 5/(c)(i)** A1

- 17.
- (a) at least 3 reflected wavefronts with same wavelength as before B1
all **and** ≥ 2 reflected wavefronts at correct angle B1
- (b) (i) no change in direction **and** clearly smaller, approximately constant wavelength B1
- (ii) 1 reduces B1
2 constant B1
- 18.
- (a) line joining points of same phase, e.g. line joining crests B1
- (b) (i) decreases B1
- (ii) decreases B1
- (iii) constant B1
- 19.
- (a) (i) up and down arrow shown
- (allow if one arrow and up/down stated in (ii)) B1
- (ii) 4 times in one second **or** once in 0.25 sec B1
- (b) $v = f\lambda$ in any algebraic form B1
- 0.8/4 C1
- 0.2 m A1
- (c) halve the frequency **or** move hand once every 0.5 springs B1
- (**ignore** move hand slower or at half speed or speed of wave double unless spring stretched more)